

A User Profile System for the Finance Platform of Commerce

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Abstract

With the rapid advancement of big data technology, user profiling has emerged as a critical tool for enterprises to refine operations and enhance precision marketing. This paper explores the application of user profiling within the financial platform of commerce, focusing on the integration of internal and external data resources, data mining techniques, and the construction of a user profile model and system. The proposed system aims to achieve precise marketing, risk control, and informed decision-making by providing comprehensive insights into user behavior and preferences. The system encompasses modules for individual and group user profiles, behavior trend analysis, and tag management, facilitating detailed user segmentation and targeted marketing strategies. A sample K-means clustering algorithm is also presented to demonstrate user grouping based on shared characteristics. The findings highlight the potential of user profiling to improve customer satisfaction and overall competitiveness in the financial industry, while promoting cross-industry data integration and the continuous evolution of user profile models.

CCS Concepts

• Information systems → Enterprise information systems.

Keywords

User Profile System, Behavior Analysis, Data Processing, Data Integration

ACM Reference Format:

Zhonghuang Hu, Yubo Qiu, Songnan Hu, Zijie Cheng, and Shuai Qiu. 2024. A User Profile System for the Finance Platform of Commerce. In *2024 the 12th International Conference on Information Technology (ICIT) (ICIT 2024)*, December 13–15, 2024, Kuala Lumpur, Malaysia. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3718391.3718436>

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ICIT 2024, Kuala Lumpur, Malaysia

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ACM ISBN 979-8-4007-1737-6/24/12

<https://doi.org/10.1145/3718391.3718436>

1 INTRODUCTION

The information on the web is always too general to solve individual user's need. An accurate representation of a user's interest is stored in a user portrait. Researchers are interested in how to identify the interest of the user and build an accurate profile.

In the year of 2015, the State Council of China proposed the "Action Plan for Promoting the Development of Big Data" [4] pointed that "the data has become a basic strategic resource for the country, and big data is increasingly becoming a new driving force for economic transformation and development." In 2016, in response to the development and application of big data, the China Banking and Insurance Regulatory Commission emphasized the need to proactively formulate a big data strategy, actively establish a big data service system, strengthen data sharing, deepen big data applications, and unleash the value of data. In view of the huge data system and massive data, developing valuable information through data mining has become an important challenge. As the Internet gradually enters the era of big data, it will inevitably bring a series of changes and reshapes to enterprises and user behaviors; the biggest change is that all user behaviors are "visualized" in front of enterprises. With the in-depth research and application of big data technology, enterprises are increasingly focusing on how to use big data to serve refined operations and precision marketing, and then further explore potential business value. Therefore, the concept of user profiling came into being. User portrait can make the service objects of products more focused and more focused. To increase client's satisfaction and increase the competition, financial institutions, especially banking institutions, need to rapidly innovate their products. To make good use of existing data assets to make data valuable. To achieve this goal, the use of user profile as a tool has become a breakthrough in solving industry pain points. User profiles can effectively present user characteristics and provide support for product innovation. Traditional financial industry relies on communication between account managers and customers to understand customers. The user profile system can effectively break the resource binding relationship between account managers and customer resources. Anyone can understand customers comprehensively and three-dimensionally through the system.

As industry competition continues to intensify, product and service homogeneity and customer loyalty decrease, the stickiness between users and products gradually decreases, and the user churn

rate becomes more serious. Therefore, it is particularly important to use multiple data sources and apply big data technology to achieve more accurate user profiles, thereby promoting product optimization, iterative upgrades of business, customized services and more accurate marketing.

The rest of this paper is organized as follows: In Section 2 we present related work. Preliminaries are given in Section 3. In Section 4, we present our user profile model and the system of user profile is introduced in section 5. We conclude the work of this paper in Section 6.

2 RELATED WORK

The Internet industry was the first to propose and apply user profile technology. The concept originated from the Persona concept proposed by Alan Cooper, the father of interactive design. Alan Cooper explained that Persona is a virtual user corresponding to a real user, and is a target customer model based on a series of real data. Based on this technology, the user profile system was developed. Initially, the University of California gradually learned by displaying and collecting users' satisfaction with the current page to establish a user interest model. Later, the California University of Management developed the user profile system Web Watcher [2], which changed from individual users to group users, using data collectors to record users' behavior, preferences, and basic information on the Internet, and using this statistical information to derive the browsing behavior of user groups, thereby building a user interest model.

Currently, the need for software systems to automatically adapt to their users has been recognized and the research on user profile [9]. A user model is initialized either with default values or by querying and grouping the user in "stereotypes" [1]. User profile has been widely applied in the financial industry, e-commerce industry, power industry, and telecom industry. Especially in the financial industry, with the popularization of big data technology and the emphasis on data in the financial industry, user profile has gradually become an important means for financial institutions to enhance their competitiveness. Recently, financial institutions integrate internal and external data resources, utilize data mining and big data analysis techniques, construct user profile models, and achieve comprehensive understanding and accurate prediction of customer behavior. This not only helps financial institutions achieve precise marketing and risk control, but also improves customer satisfaction and loyalty.

Franca *et al.* did research in the field of psychology highlighted the value of identifying users' personality traits, which helps build adaptive and personalized systems and improve user experiences [5]. Gu *et al.* proposed a psychological modeling method based on computational linguistic features to profile personality traits of users on Sina Weibo and their correlations with user's social behaviors [6]. This research investigates the potential relationship between personality traits of users, profile information, and social-network behaviors on Sina Weibo. This approach understood demographic and psychological profiles of users on social media without customer disruption. Chen *et al.* used classification algorithms in machine learning to predict users' demographic attributes based on a month of data collected with the Sogou search engine in order to generate users' profiles [3]. The proposed model is a

two-stage structure using a month of data with demographic labels as the training data. The first stage of the structure is based on traditional machine learning models and neural network models, whereas the second one is a combination of the models from the first stage. Paper [7] mines mobile user behavior patterns and constructs a group user profile by analyzing access data collected from the WeChat Mini Program. They extract the attributes of mobile users considering their age group, geographic information, and on-line duration. Using Z-score standardized processing and K-means clustering algorithm, they model the user profiles through three dimensions including interaction intensity, daily average duration, and access frequency. Zhang *et al.* constructed a power user behavior assessment model based on a huge number of user profiles [11]. After text conversion, data cleaning and variable attribute analysis the huge number of grid customer service phone data, a lot of user profiles are established. The logistic regression model is combined with the decision tree algorithm to complete the construction of the electric power user behavior evaluation model.

The e-commerce industry utilizes the personal profile function of social platforms for marketing and promotion, thereby enriching the data dimensions of user consumption habits, consumption ability, consumption preferences, etc. accumulated by the e-commerce industry itself, and efficiently conducting marketing and promotion [10]. With the rapid development of big data analysis technology and the rapid accumulation of data in various industries, user profiles are gradually utilizing massive amounts of data between industries to promote active understanding of users and provide users with better experiences [8]. The enrichment of data dimensions and the breaking of barriers to information exchange between industries have driven the development of data analysis and the improvement of user profile models, thus forming a virtuous cycle mechanism. Methods and applications.

3 PRELIMINARY

3.1 Data Acquisition and Data Processing

Data acquisition is based on varying user participation methods, the collection of raw user profile data can be categorized into two approaches: explicit and implicit acquisition. Explicit acquisition involves users manually inputting their basic information, behavioral preferences, and related data. This straightforward method offers high-quality data but necessitates active user engagement and typically limits the scope of accessible information. Conversely, implicit acquisition is transparent to users and encompasses two main techniques: logging user information during interactions and analyzing user behavior patterns to deduce characteristics. The richer data obtained through implicit methods results in more comprehensive user profiles. In practice, both methods are often combined by industry and academia to leverage the diverse data trails left by users.

Data processing refers to the purification of raw data to eliminate impurities, yielding standardized data suitable for constructing user profiles. Key data processing techniques include word segmentation, data filtering, and data normalization. Word segmentation is crucial for unstructured text data processing; data filtering ensures the reliability of basic data by ridding it of impurities; and data normalization maintains consistency across input model data.

3.2 Tag Selection and Tag Weight

Tags symbolize specific user characteristics and serve as a highly correlated content organization method, enabling swift identification of suitable content and classifications. From a computational hierarchy perspective, tags can be categorized into fact tags and model tags. Fact tags are derived from statistical analyses of raw data, for instance, user purchase frequency, which reflects a statistical overview of a user's actual purchase behavior over a specified duration. Conversely, model tags are formulated based on fact tags by establishing associations with business challenges, thereby yielding tags tailored for user profile selection models.

Tags primarily address the issue of description. However, in practical applications, resolving the problem of data correlation is equally important. Hence, tags are often designed as a comprehensive system encompassing multiple tags. Each tag holds varying degrees of significance in portraying specific dimensions of a user profile. Consequently, assigning weights to these tags based on their importance in characterizing user profiles is imperative. The prevalent weighting methodologies can be broadly classified into subjective and objective weighting methods. Objective weighting methods encompass the entropy weight method, correlation coefficient method, and standard deviation method, among others. Subjective weighting methods include the subjective experience method, expert survey weighting method, Delphi method, and analytic hierarchy process.

3.3 User Profile Model selection

User profiling technology is a multidisciplinary combination that requires knowledge fusion in areas such as knowledge graphs, natural language processing, machine learning, and data mining. The commonly used user profile models are mainly divided into three categories: text mining techniques, classification algorithms, clustering algorithms, etc. Text mining mainly focuses on the mining and processing of unstructured data, including TF-IDF, vector space models, topic models, etc; Classification algorithms are suitable for supervised learning user classification, including artificial neural networks (ANN), proximity algorithms (KNN), support vector support machines (SVM), etc; Clustering algorithms are suitable for unsupervised learning user clustering, including k-means clustering algorithm, k-centroid algorithm, density based clustering algorithm DBSCAN, hierarchical agglomerative clustering algorithm HAC, etc.

The application scope of user profiling is very broad, mainly including three aspects: firstly, by analyzing potential users and targeting specific user groups for advertising placement, unnecessary cost investment can be reduced; Secondly, by analyzing the correlation of user data, a personalized recommendation system for users can be constructed to achieve customized deployment of services or products for thousands of people; Thirdly, evaluate the operational effectiveness of the enterprise, improve product operations, enhance service quality and user experience, and promote the healthy development of the enterprise.

4 THE USER PROFILE MODEL

To build a user profile model, usually it requires three steps: (1) Training the classifier. The training data for a particular concept are

merged as a super-document to train the classifier for each concept. A super-document goes through an indexing to calculate a vector. The weight of each vocabulary term (vector) is stored in that concept. Thus, each concept is treated as n-dimensional vectors in which n represents the number of unique terms in the vocabulary. Each term weight is calculated as $tf \times idf$ and normalized by their length. In more detail, uw_{ij} represents the unnormalized weight of term i in concept j , is calculated as equation (1) and (2)

$$uw_{ij} = t_{fij} \times idf_i \quad (1)$$

$$idf_i = \log \frac{\#ofdocumentsinthe\ collection}{\#ofdocumentsinthe\ collection\ that\ contain\ t_i} \quad (2)$$

where t_{fij} is the number of occurrences of t_i in sd_j , idf_i is the super-document used for training concept j .

(2) Collecting user data. The documents are stored in a log file by a proxy server. The documents are extracted for each user and filtered to remove documents that are considered as not suitable [10].

(3) classifying the documents. Documents and concepts are listed in the vector model and the similarity is calculated in the vector. The similarity between concept c_j and document d_k is calculated as equation (3).

$$similarity(c_j, d_k) = c_j \circ d_k = \sum_{i=1}^n w_{ij} \times d_{jk} \quad (3)$$

where n represents the number of terms in the vocabulary, w_{ij} is the normalized weight or term i in concept j .

After the similarity is calculated, the results are sorted, the weights from documents which match the same concept are accumulated. A user file consists all concepts with weights are created as a result.

As shown in Figure 1, the profile model contains four layers:

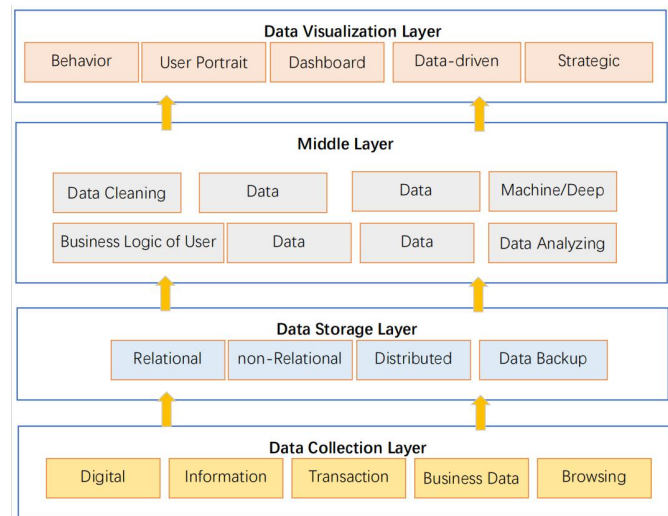


Figure 1: The User Profile Model

Data Collection Layer: This foundational layer serves as the cornerstone and indispensable prerequisite for building user profiles. It involves the systematic gathering of a myriad of data points

from a variety of sources. These sources include but are not limited to user interactions on digital platforms, user information, transactional histories, browsing behaviors, business activities, and more. The data collected is both structured (e.g., numerical data in databases) and unstructured (e.g., text data from social media posts). The quality, completeness, and timeliness of the data collected are crucial factors that directly influence the accuracy and reliability of the user profiles generated.

Data Storage Layer: This layer is responsible for securely storing and managing the vast amounts of data collected from the data collection layer. It employs advanced database management systems and storage solutions to ensure the data is organized, accessible, and scalable. The data storage layer also supports data backup and recovery to ensure that data can be restored quickly and efficiently in the event of a failure. Integrity checks, such as data validation and consistency enforcement, are performed to maintain data accuracy and reliability. Effective management of the data stored in this layer enables seamless access and retrieval for subsequent processing and analysis in the Middle Layer.

The Middle Layer: Positioned between the data storage layer and the data visualization layer, this layer is a multifaceted entity that can be both abstract and tangible. It serves as the brain of the profile system, interacting with both the data storage layer and the data visualization layer to implement the intricate business logic of user profiles. The middle layer is tasked with processing, integrating, and analyzing the data stored in the bottom layer. This includes data cleaning to remove inconsistencies and errors, data normalization to ensure uniformity, and data enrichment to add context and value. Additionally, the middle layer employs advanced algorithms and machine learning models to extract meaningful insights and patterns from the data. These insights are then used to create detailed and nuanced user profiles that encapsulate various aspects of user behavior, preferences, demographics, and more. Through sophisticated data visualization tools and systems, the data results and models implemented in the middle layer can be transformed into visually compelling and easily understandable formats. This enables business users and analysts to gain insights into user behavior, identify trends, and make informed decisions based on data-driven insights.

Data Visualization Layer: This is the pinnacle of the profile system, often referred to as the narrow user profile. It represents the final output, presenting the insights and patterns derived from the middle layer in a visually appealing and intuitive manner. The data visualization layer employs a variety of charts, graphs, dashboards, and other visual elements to display user profiles, trends, and patterns. These visualizations enable business users to quickly and easily understand complex data sets, identify key trends and insights, and make data-driven decisions. By providing a clear and concise view of user behavior and preferences, the data visualization layer supports strategic planning, product development, marketing campaigns, and more.

5 A USER PROFILE SYSTEM

The user profile system is developed to provide precise marketing, risk prevention and control, decision-making. The main functionalities currently contain three parts, as shown in Figure 2:

User Profile Module includes three main functionalities. **List of User Profiles** contains a collection of all user profiles created within the system, allowing for easy access and overview. **Individual User Profile** details profile of individual user, including user basic information, user financial information, user working information, user company information, user preferences, purchase history, and any other relevant data points. This provides a holistic view of each user. **User Behavior Trends** provide analysis and visualization of user behavior over time, identifying patterns, trends, and changes in user preferences or interactions. This can help predict future behaviors and optimize user engagement strategies.

User Group Profile Module include five major functionalities. **List of User Groups Profiles** contains a collection of all user group profiles, enabling quick navigation and comparison between different user groups. **Individual User Group Profile** contains in-depth profiles of a specific user group, detailing shared characteristics, interests, and behaviors. This helps tailor marketing and product development efforts to better meet the needs of these groups. **User Group Analysis** details examination of user group data, including demographics, preferences, and behaviors, to gain insights into the unique qualities and challenges of each group. **User Group Trends** provide tracking and analyzing trends within user groups, such as changes in preferences or emerging behaviors, to stay ahead of market shifts and user needs. **User Group Company Analysis** evaluates how user groups interact with specific companies, identifying areas for improvement, and opportunities for deeper engagement or loyalty building.

Tag Information Module contains three key functionalities. **List of User Tags** includes a list of all tags assigned to individual users, helping to categorize and group users based on shared attributes and behaviors. **List of User Group Tags** is similar to user tags, but for user groups, allowing for the categorization and analysis of groups based on common characteristics and trends. **Tag Management** provides tools for creating, editing, and deleting tags, as well as assigning them to users or user groups. This feature ensures that the tagging system remains up-to-date and relevant, supporting accurate and efficient user segmentation and analysis.

5.1 Sample Characteristics of User Profile

The Table 1 describe a list of sample characteristics of user's basic information used in profile.

5.2 A Sample K-means User Grouping Algorithm

The K-means algorithm is an unsupervised learning algorithm used to divide a dataset into K clusters (groups), where each cluster is represented by its centroid (center point) [8]. The algorithm continuously updates the centroid and grouping through iteration until the convergence condition is reached. In user grouping applications, users with similar characteristics or behaviors can be grouped together to facilitate subsequent recommendations, analysis, and other applications.

A sample K-means algorithm for grouping users is provided as follows.

- (1) Initialize centroids: Randomly select K users from the user dataset as the initial centroids.

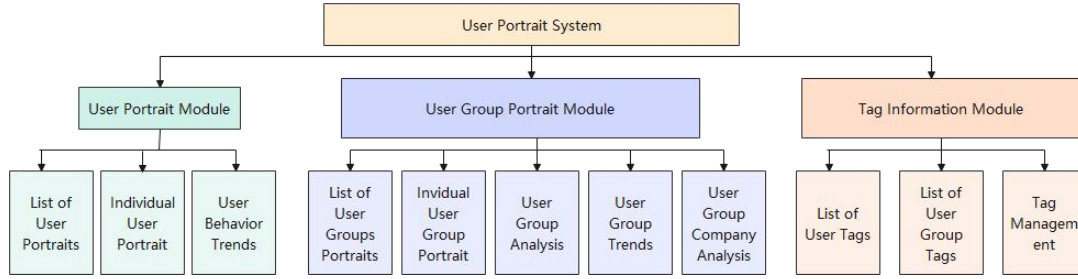


Figure 2: Functionalities of the User Profile System

User Info	User Financial Info	User Working Info	User Transaction Info
Name	Contribution	Company	Transaction of Parties
Age	Level of Credit	Position and Rank	Transaction Items
ID	Income	Work Experience	Time and Location
Education	Expenditure	Year of working in the Company	Transaction Status
Gender	Row Asset	Industry Background	Price and Payment Method
Email	Debt		Relevant Information
Cell Phone			Transaction History

Table 1: User Characteristics

- (2) Iteration process: Start iterating until the maximum number of iterations $Max_iteration$ is reached or the change in centroid is less than the threshold $Threshold$.
- (a) Grouping: For each user, calculate their distance to all centroids and assign them to the group with the centroid with the smallest distance. During initialization, set a maximum value as the minimum distance, traverse all centroids, find the centroid with the smallest distance, and record the index of that centroid as the user's group ID. After assigning each user to the corresponding group, the current grouping result $User_Groups$ is obtained.
- (b) Update centroid: For each group, calculate the centroid of all users within the group (i.e., the new centroid). This is usually achieved by calculating the average of all user information within the group. Store all new centroids in the $new_centroids$ list.
- (3) Check convergence: If the change between the new and old centroids is less than the given threshold $Threshold$, the algorithm is considered to have converged and the iteration is stopped. Otherwise, assign the new centroid to the original centroid and continue with the next iteration.
- (4) Return result: When the iteration ends, return the final grouping result $User_Groups$ and centroid $Centroids$.

6 CONCLUSION

User profiling provides comprehensive user insights for financial institutions through in-depth analysis of user behavior and preferences. This not only helps financial institutions achieve precise marketing and risk control, but also significantly improves customer satisfaction and loyalty. In the financial supply chain, user profiling has become a key means for financial institutions to enhance their competitiveness. With the increase of data dimensions

Algorithm 1 K-means Algorithm for Grouping Users

Input: $User_Dataset$, K , $Max_Iteration$, $Threshold$

Output: $User_Groups$, $Centroids$

```

1: for iteration in range( $Max\_Iteration$ ) do
2:    $User\_Groups = []$ 
3:   for user in  $User\_Dataset$ : do
4:      $min\_distance = \text{maximum value}$ 
5:     for centroid in  $Centroids$ : do
6:        $distance = \text{calculating the distance between user and centroid}$ 
7:       if  $distance < min\_distance$ : then
8:          $min\_distance = distance$ 
9:          $group\_id = \text{current centroid's index}$ 
10:      end if
11:       $User\_Groups.append(group\_id)$ 
12:   end for
13: end for
14:  $new\_centroids = []$ 
15: for  $i$  in range  $K$ : do
16:    $Group\_Users = \text{select all users in } User\_Dataset \text{ in group } i$ 
17:    $new\_centroid = \text{calculating } Group\_Users's \text{ center}$ 
18:    $new\_centroids.append(new\_centroid)$ 
19: end for
20: if the change of group center is less than  $Threshold$  then
21:   Break
22: end if
23:  $Centroids = new\_centroids$ 
24: end for
25: return  $User\_Groups$  and  $Centroids$ 
  
```

and the breaking of information exchange barriers between industries, the construction of user profiles will pay more attention to the integration and sharing of cross-industry data. This not only enriches the content of user profiles, improves their accuracy and comprehensiveness, but also promotes the development of data analysis technology and the improvement of user profile models, forming a virtuous cycle mechanism.

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